

PROJECT BOATY

Design FMEA

Failure modes and effects analysis of the Mk1 architecture, used to drive simulator, rig and bench tests

Document	BOATY-FMEA-001
Issue	Issue C (for review)
Date	28 September 2026
Status	Design stage (before hardware). Updated after rigs, pool and before the first lake trial (SAF-006).
Basis	ADD Issue C, ICD Issue B, SSS Issue A/B
Content	48 failure modes, 19 actions, 68 catalogued tests

Document control

Issue	Date	Change	By
A	28 September 2026	First issue: design FMEA at architecture level.	Claude (drafted)
B	28 September 2026	Actions carried into SSS-HLM/MCP/MCN Issue B, ICD Issue C and ADD Issue D; status and post-action ratings added. A-05, A-09, A-17 await the operations manual.	Claude, owner request
C	28 September 2026	CR-03: salvaged cells (FM-23 O 1 → 2; A-20 cell acceptance) and adult PIN (new FM-48). Operations manual BOATY-OPS-001 closes A-05, A-09, A-11, A-17, A-19.	Claude, owner decision

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: challenge the ratings, especially occurrence, which is a judgement before any hardware exists. And look for missing failure modes: an FMEA is only as good as its imagination.

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1. Purpose and scope

This design FMEA looks for ways Boaty Mk1 can fail and asks three questions of each. How bad is the effect? How likely is it? Will our planned tests find it before the boat goes on the lake? It works at the level of the ADD's subsystems and interfaces, before detailed design. It has two jobs:

- **Drive the tests.** Every failure mode with severity ≥ 8 must be exercised by a simulator, rig or bench test in the SSS-SIM catalogue. The build checks this.
- **Change the design early.** Where a risk is high, actions go into the next subsystem-spec issues while changes are still cheap.

It supports SAF-006 (FMEA before the first lake trial) and FS-013 (no single failure leaves the fence under power). It is a living document: ratings are revisited with rig, pool and lake evidence.

2. Method

2.1 Rating scales

S	Severity (effect)	O	Occurrence	D	Detection by planned verification
10	Injury to a person, or fire	8	Expected most sessions	2	Certain: an automated test exercises it directly
9	Boat leaves the inclusion fence under power, or risk of collision with a person	6	Frequent: several times a season	3	High: a scripted scenario or rig test targets it
8	Boat unrecoverable, or wildlife harmed or disturbed	5	Occasional	5	Moderate: inspection, analysis or an indirect test
7	Boat stops or drifts inside the fence; manual recovery needed	4	Likely during Mk1's life	7	Low: likely to show only in pool or lake use
6	Uncommanded behaviour, contained; boat returns	3	Possible, but unlikely	9	Very low: unlikely to be found before use
5	Mission aborted; boat returns home safely	2	Remote	10	None
4	Mission degraded (no photos, no plan → templates)	1	Practically nil		
3	Minor inconvenience (reboot, re-plan)				
2	Cosmetic or record gap				
1	No effect				

2.2 Rules

- Action required if $S \geq 9$ (whatever O and D are)
- Action required if $RPN = S \times O \times D \geq 100$
- Action required if $D \geq 7$ and $S \geq 7$
- Every row with $S \geq 8$ must have at least one test

D rates our verification (will we find it before the lake?), not detection in operation. In-operation detection is listed separately as a control.

3. FMEA worksheet

Blue S cell = severity ≥ 9 . Orange RPN cell = at or above the action threshold. 'After' = RPN once the incorporated actions are in (Issue B); '-' = unchanged. Test IDs are defined in SSS-SIM Issue B, section 6.

ID	Item / function	Failure mode	Effect	S	Cause	O	Controls: prevent / detect in use	D	RPN	After	Tests	Actions
FM-01 HLM	GNSS position	No fix	Motors stop (FS-004); boat drifts inside the fence	7	Sky blocked, module or cable fault	3	Mast mounting; M10 multi-constellation / EKF failsafe	3	63	-	SC-04, L2-03	-
FM-02 HLM	GNSS position	Position jump (glitch)	False breach, or wrong steering near the fence	9	Multipath off water, reflections	4	EKF glitch rejection / EKF innovation checks	5	180	108	SC-20	A-01
FM-03 HLM	GNSS position	Frozen but reported valid	Helm navigates blind; could leave the fence	9	Receiver hang, data line fault	2	GNSS data-age checks / EKF failsafe	5	90	54	SC-21, L2-04	A-01
FM-04 HLM	Heading	Compass disturbed by motor current	Curved tracks or circling	6	Current loops near the compass	6	Mast mount ≥ 150 mm; interference calibration / EKF yaw checks; GNSS-velocity yaw fallback	3	108	72	SC-22, L2-05	A-02
FM-05 HLM	Heading	Compass orientation wrong	Boat drives the wrong way; RTL also wrong; persistent breach	9	Parameter or mounting error	3	Parameter baseline check / EKF consistency once moving	3	81	54	SC-22, SC-38, L2-06	A-03
FM-06 HLM	State estimate	EKF diverges	Erratic control; failsafe HOLD	7	Sensor fault, vibration	2	Stable firmware, mounting / EKF failsafe	3	42	-	SC-04	-
FM-07 HLM	Flight controller	Firmware crash / reboot mid-mission	Outputs stop; boots disarmed; drifts	7	Firmware fault	2	Pinned stable release / Heartbeat loss alarm	5	70	-	SC-23, L2-07	-
FM-08 HLM	Flight controller supply	Brownout on thrust step	Reboot as FM-07	7	Inadequate supply design	3	PWR-D10/D12 / Logs	3	63	-	L2-08	-
FM-09 HLM	Configuration	Parameters differ from baseline	A failsafe doesn't fire as specified	9	Stale or edited parameters	4	Baseline file (SAF-007) / Mismatch blocks arming (MCN-D45)	2	72	-	SC-24	-
FM-10 HLM	Fence	Fence not loaded or disabled	No fence protection	9	Upload skipped or failed	3	Pre-arm fence check / Arming refused	2	54	-	SC-25	-

ID	Item / function	Failure mode	Effect	S	Cause	O	Controls: prevent / detect in use	D	RPN	After	Tests	Actions
FM-11 MCN	Site data	Coordinates swapped or wrong site	Fence in the wrong place	9	GeoJSON [lon, lat] order error, file mix-up	3	Site linter / Pre-arm: home inside fence	3	81	54	SC-26	A-04
FM-12 MCN	Site data	Fence doesn't match the real shore (imagery offset)	Boat reaches bank or reeds inside its 'fence'	7	Satellite image offset, drawing error	5	5 m margin / Operator watching	7	245	70	R-01	A-05
FM-13 HLM	RTL path	RTL crosses an exclusion zone	Island or nesting area approached	8	No path planning on board	3	Fence-aware planning (V-01) / Exclusion breach → stop	3	72	-	SC-27	-
FM-14 HLM	Fence	Persistent breach doesn't stop the motors	Boat keeps pushing at the shore	8	No native mechanism (V-14)	3	Interim HOLD from MCP and Mission Control / Breach alarm	5	120	72	SC-28	A-18
FM-15 HLM	Battery state	Charge over-estimated	Late RTL; dies before home; drifts	7	Wrong capacity, sensor calibration	4	Calibration (PWR-D13) / mAh failsafe only	3	84	42	SC-39, L2-09	A-06
FM-16 HLM	Stuck detection	Doesn't trigger in weed	Motors keep driving into weed; battery drains	7	Boat creeps above the speed threshold	5	Crash check (V-05) / Battery failsafe eventually	5	175	84	SC-05, L3-02	A-07
FM-17 PRP	Thruster	One motor or ESC dead or fouled	Boat circles; can't navigate or RTL	7	Motor failure, weed, connector	4	Keyed connectors, guards / None native	7	196	84	SC-29	A-08
FM-18 PRP	Motor wiring	Direction or channel reversed	Spins or drives backwards on the first command	9	Build error	4	IF-05 mapping / First-motion check (A-03)	2	72	54	L2-01, SC-38	A-03
FM-19 PRP	ESC	Doesn't stop on signal loss	Runaway if the helm dies	9	ESC configuration	3	ESC failsafe setting / -	2	54	-	L2-10	-
FM-20 PRP	ESC	Restarts at non-neutral when signal returns	Sudden jolt	6	ESC configuration	3	Neutral re-arm / -	3	54	-	L2-10	-
FM-21 PRP	Prop guard	Guard damaged	Finger injury risk at recovery	10	Impact, handling	2	8 mm probe design / Checklist inspection	5	100	50	B-01	A-09
FM-22 PWR	Key switch	MOSFET fails short (rail always live)	Latent: arming protection is software-only	6	Component failure	2	Rated part / Checklist rail test (A-10)	5	60	36	L2-02, L2-13	A-10

ID	Item / function	Failure mode	Effect	S	Cause	O	Controls: prevent / detect in use	D	RPN	After	Tests	Actions
FM-23 PWR	Battery	Thermal runaway or fire in the box	Fire, injury	10	Cell damage, short, BMS fault; salvaged cells (Issue C: O 1 → 2)	2	BMS, fuse, hard cells, off-boat charging / None in operation	7	140	50	B-02, B-07	A-11, A-20
FM-24 HUL	Box sealing	Water ingress	Shorts: sudden stop or erratic behaviour	7	Gland or lid seal	4	Glands, dunk test / Moisture sensor → RTL	3	84	-	SC-10, B-03	-
FM-25 PWR	Harness	Fuse blows or connector pulls	Total power loss; drifts	7	Overload, vibration	2	Rated fuse, strain relief / Heartbeat loss	3	42	-	B-04	-
FM-26 MCP	Mission computer	Crash or SD corruption	No photos; B4-B6 absent; mission still ends in RTL	5	Power cut, SD wear	4	Read-only root / Mission Control sees MCP silent	2	40	-	SC-07, L1-08	-
FM-27 MCP	Boat services	Wrong command sent (bug)	Unwanted mode change; stuck in GUIDED	6	Software bug	3	Command filter; bounded GUIDED (V-11) / Helm mode visible	3	54	-	SC-06, SC-30	-
FM-28 MCP	UART link	Traffic floods the UART	GCS heartbeats delayed → false link-loss HOLD	5	Stream rates too high	3	Rate budget (IF-04) / Latency logging	3	45	-	L1-05	-
FM-29 MCP	Camera	Camera fails	No photos	4	Ribbon, module	3	Strain relief / Health: camera_ok	2	24	-	B-05	-
FM-30 MCP	Clock	Wrong time before GNSS	Photo and log times wrong	2	No RTC, no internet	5	GNSS time sync / Flagged timestamps	3	30	-	B-06	-
FM-31 MCN	Mission Control	Crash or power bank dies mid-mission	No UI, no STOP; boat returns via B4	5	Software, battery	4	Power budget, autostart / Helm GCS failsafe	2	40	-	SC-03, L1-06	-
FM-32 MCN	Panel	STOP button fails	Can't stop from the panel (web STOP remains)	7	Wiring, switch	2	Quality switch / Checklist button test (A-13)	3	42	21	L1-02	A-13
FM-33 MCN	Panel	GO pressed by accident	Mission starts before the adult is ready	6	Child curiosity	4	1 s hold; adult arming / State shown and spoken	2	48	-	L1-02	-
FM-34 MCN	Validator	Accepts a mission crossing a boundary	Exclusion approached; helm fence still protects	8	Software bug	2	95% branch coverage / Helm fence	2	32	-	SC-31	A-14

ID	Item / function	Failure mode	Effect	S	Cause	O	Controls: prevent / detect in use	D	RPN	After	Tests	Actions
FM-35 MCN	Read-back	Mismatch not detected	Uploaded mission differs from approved	7	Canonical-form bug	2	Contract tests / -	2	28	-	SC-32	-
FM-36 MCN	Planner	Plausible but unintended plan approved	Longer or different mission than wanted	4	Adult approves without reading	5	Spoken summary, map / Visible on the map	5	100	60	SC-33	A-15
FM-37 MCN	Planner	Odd or adversarial instruction	Request to chase ducks etc.	6	Child phrasing, injection	3	Intent vocabulary, site names only / Validator	3	54	-	SC-33	-
FM-38 MCN	Internet	No internet	Templates only	3	No mobile signal	6	Offline templates / UI message	2	36	-	P-01	-
FM-39 MCN	Link	Link flaps at the range edge	Repeated HOLD/continue in MANUAL	5	Range, geometry	6	Pole antenna / RSSI display	3	90	-	SC-34, L1-07	-
FM-40 MCN	Link	Commands delayed > 1 s	STOP late	7	Congestion, interference	3	Traffic budget / Latency logging	3	63	-	SC-34, L1-03, L2-11	-
FM-41 MCN	GCS identity	Two ground stations connected	Conflicting commands	6	QGC left running	3	MC-014 procedure / -	5	90	36	SC-37	A-16
FM-42 OPS	Launch	Launched with wind away from the bank	A dead boat drifts away	7	Procedure not followed	4	Checklist / Operator	5	140	70	R-01	A-19
FM-43 OPS	Other users	Swimmer or paddler near the boat	Collision risk (slow boat, guarded props)	9	Shared water	3	OPS-003/004, low speed, guards / Operator lookout	7	189	126	R-01	A-17
FM-44 OPS	Wildlife	Nesting area approached	Disturbance	8	Exclusions missing	2	OPS-005 exclusions / Operator	3	48	-	R-01, SC-27	-
FM-45 OPS	Handling	Child handles a pod with the key in	Prop injury risk	10	Supervision lapse	2	Guards; adult-only arming / Key-out rule (A-09)	5	100	50	B-01	A-09
FM-46 MCN	Home	Home set wrongly	RTL goes to the wrong place	6	Armed before GNSS settled	3	Home shown before GO / Map	3	54	-	SC-35	-
FM-48 MCN	Adult unlock	Child learns or guesses the PIN (Issue C)	Could arm or approve without an adult. Motors still need the magnetic key.	6	PIN seen or shared	3	6 digits, lockout, never shown or spoken; the magnetic key physically gates the motors / Checklist; approval log	3	54	-	L1-10	-

ID	Item / function	Failure mode	Effect	S	Cause	O	Controls: prevent / detect in use	D	RPN	After	Tests	Actions
FM-47 MCN	Mission upload	Upload interrupted (partial mission)	Wrong mission executed	6	Link drop mid-transfer	4	Transfer protocol / Read-back (VAL-010)	2	48	-	SC-36, SC-32	-

4. Results

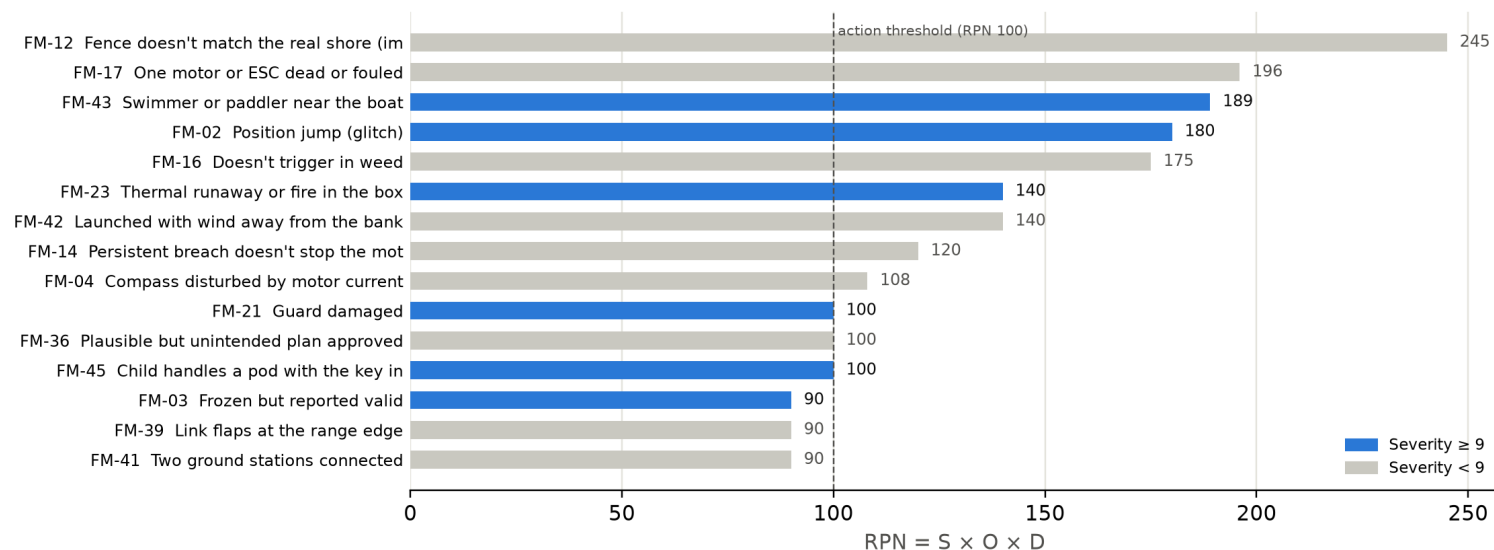


Figure 1. The 15 highest RPNs. Severity ≥ 9 rows need action or a direct test whatever their RPN.

Severity profile: 3 rows at S10, 9 at S9, 4 at S8; 16 rows at S ≥ 8, all with tests. 19 rows meet an action rule. Those with a direct test (D ≤ 3) are covered by the test; the rest have actions.

4.1 What the FMEA changed

- **Three new boat-side safeguards** (actions A-03, A-07, A-08). A first-motion heading check catches a reversed compass or motor. A second stuck detector covers weed creep. A navigation-divergence watchdog catches a dead motor. All three only request HOLD, so they stay within the ADD's rule for boat-side Python.
- **Procedures matter as much as code.** The fence-vs-shore offset (A-05), the launch point for the wind (A-19), the lookout for other water users (A-17) and key-out handling (A-09) come from operational failure modes. They go into an operations manual.
- **Twenty FMEA-derived simulator scenarios (SC-20 to SC-39)** and thirteen iron-bird rig tests. The first catalogue only covered the failsafes we'd already designed.

4.2 After Issues B and C

Every action is now incorporated: in the subsystem specs (Issue B/C) or in the operations manual (BOATY-OPS-001). The one row still above the action threshold is FM-43 (other water users near the boat, RPN 126 after actions). It's a severity-9 hazard with no sensor to detect it, so detection relies on the

lookout procedure (OP-10). It remains an accepted residual risk (section 6). Highest RPNs now: FM-43 126, FM-02 108, FM-39 90, FM-16 84, FM-17 84, FM-24 84.

5. Actions

ID	Action	Owner	Failure modes	Status (Issue B)	Carried by
A-01	Confirm GNSS glitch and freeze protection parameters; add SC-20/21	HLM	FM-02, FM-03	Incorporated	HLM-D39
A-02	Measure compass interference on rig L2 before the hull layout is frozen	HLM	FM-04	Incorporated	HLM-D41
A-03	Add a first-motion heading check: in the first 10 s of any driven mode, GNSS course vs heading differs > 45° → HOLD + alarm	MCN / MCP	FM-05, FM-18	Incorporated	MCP-D22, MCN-D60
A-04	Site linter: home inside the fence; site within 1 km of the configured reference; coordinate-order sanity	MCN	FM-11	Incorporated	MCN-D53
A-05	Fence survey procedure: walk the launch bank with a GNSS reference to check imagery offset before the first mission at a site	OPS	FM-12	Incorporated	OP-01
A-06	Add voltage thresholds as a backstop to the mAh battery failsafe	HLM	FM-15	Incorporated	HLM-D40
A-07	Second stuck detector in the MCP: throttle high and progress along track low for 10 s → HOLD	MCP	FM-16	Incorporated	MCP-D23
A-08	Navigation-divergence watchdog in the MCP: cross-track > 10 m or heading error > 60° for 20 s → HOLD + alarm	MCP	FM-17	Incorporated	MCP-D24
A-09	Checklist and procedure: key out before anyone touches the boat; lift only by the handle or hoop; inspect guards	OPS	FM-21, FM-45	Incorporated	OP-05, OP-09, OP-11, CL-09
A-10	Checklist step: key out → confirm motor rail reads 0 V on screen	MCN	FM-22	Incorporated	MCN-D54
A-11	Box temperature sensor with alarm and RTL above 60 °C; battery inspection before each session	MCP / OPS	FM-23	Incorporated	MCP-D26, OP-04, CL-07
A-13	Checklist button test: each crew button pressed, LED and sound confirm	MCN	FM-32	Incorporated	MCN-D55
A-14	Property-based validator tests against an independent geometry implementation	SIM	FM-34	Incorporated	SIM-D09
A-15	Plan preview shows duration and distance prominently before approval	MCN	FM-36	Incorporated	MCN-D56
A-16	C7 detects a foreign system-255 heartbeat and refuses to operate	MCN	FM-41	Incorporated	MCN-D57
A-17	Operator is the dedicated lookout; brief 'STOP if in doubt'; no missions while people are within 50 m of the planned route	OPS	FM-43	Incorporated	OP-10, CL-03, CL-17, K-06
A-18	Resolve V-14; if there's no native mechanism, specify the interim HOLD path formally	HLM	FM-14	Incorporated; V-14 still open	HLM-D24, MCN-D59
A-19	Site file lists the preferred launch point per wind direction; the checklist asks 'is the wind blowing towards us?' and suggests the launch point	MCN / OPS	FM-42	Incorporated	MCN-D58, OP-06, CL-06

ID	Action	Owner	Failure modes	Status (Issue B)	Carried by
A-20	Salvaged-cell acceptance tests and records (capacity, internal resistance, self-discharge, matching, visual)	PWR	FM-23	Incorporated	PWR-D20, PWR-D21

A-12 is unused: it was merged into A-13 during drafting. Actions going into 'SSS-x Issue B' are carried into those specification issues next.

6. Residual risks accepted for Mk1

Ref	Residual risk and rationale
FM-43	Swimmer or paddler near the boat. No sensor detects people. Relies on procedure, low speed, guarded props and the operator lookout. Accepted for Mk1 with A-17.
FM-31	If Mission Control dies, the Wi-Fi access point dies with it, so there's no remote STOP. The boat returns by itself (B4 at 60 s). Accepted: severity 5.
FM-23	Battery fire can't be detected early on board. Mitigated by hard cells, BMS, fuse and off-boat charging (occurrence 1), plus A-11. Accepted.
Common cause	The fence and the failsafes both run on the flight controller, so a flight-controller logic fault is a common cause. Mitigations: a pinned stable firmware, ESC stop on signal loss, the shore physically bounds the lake, a short mission time, and operator STOP. Accepted for Mk1. A future independent guardian (SAF-005) would remove it.

7. Test coverage of severe failure modes

FM	S	Failure mode	Tests (kind)
FM-21	10	Guard damaged	B-01 (BENCH)
FM-23	10	Thermal runaway or fire in the box	B-02 (BENCH), B-07 (BENCH)
FM-45	10	Child handles a pod with the key in	B-01 (BENCH)
FM-02	9	Position jump (glitch)	SC-20 (SIM)
FM-03	9	Frozen but reported valid	SC-21 (SIM), L2-04 (L2)
FM-05	9	Compass orientation wrong	SC-22 (SIM), SC-38 (SIM), L2-06 (L2)
FM-09	9	Parameters differ from baseline	SC-24 (SIM)
FM-10	9	Fence not loaded or disabled	SC-25 (SIM)
FM-11	9	Coordinates swapped or wrong site	SC-26 (SIM)
FM-18	9	Direction or channel reversed	L2-01 (L2), SC-38 (SIM)
FM-19	9	Doesn't stop on signal loss	L2-10 (L2)
FM-43	9	Swimmer or paddler near the boat	R-01 (REHEARSAL)
FM-13	8	RTL crosses an exclusion zone	SC-27 (SIM)
FM-14	8	Persistent breach doesn't stop the motors	SC-28 (SIM)
FM-34	8	Accepts a mission crossing a boundary	SC-31 (SIM)
FM-44	8	Nesting area approached	R-01 (REHEARSAL), SC-27 (SIM)